Adarsh Singh Tomar

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Experience

WNS Global Services — Mar 2025 – Jun 2025
Research Intern, Generative AI & RAG — Bangalore, India

- Researched and implemented RAG techniques — Vector RAG, Corrective RAG, and Graph RAG — for enterprise-scale knowledge retrieval systems serving multiple business units.
- Designed and integrated graph-based retrieval pipelines using AWS Neptune, Neo4j, and OpenSearch, enabling structured semantic search over large knowledge graphs.
- Built and deployed interactive Q&A applications using LangChain, Groq, and Streamlit with graph querying via SPARQL and Gremlin for real-time enterprise use.
- Explored Vibe Coding (Cursor-based AI assistant) by building production-style applications; conducted comparative analysis against low-code/no-code platforms, identifying benefits, limitations, and real-world applicability.
- Documented research findings, pipeline architectures, and design decisions for team handoff and internal knowledge sharing across engineering teams.

Projects

Computer Vision Powered Image Search Engine — *Python, YOLO, OpenCV, Streamlit*

- Built an end-to-end computer vision system using YOLO (Ultralytics) to detect objects in images and auto-generate searchable metadata with object labels and confidence scores.
- Implemented metadata-driven image search using logical filters (AND/OR) and object count thresholds, enabling precise multi-condition queries across large image datasets.
- Designed an interactive Streamlit UI with bounding box visualizations; modularized inference and configuration components for scalability and maintainability.

End-to-End MLOps Pipeline – Network Intrusion Detection — *Python, MLflow, FastAPI, Scikit-learn*

- Built a fully modular ML-based network intrusion detection system to classify network traffic as normal or malicious using structured feature engineering and preprocessing pipelines.
- Tracked experiments with MLflow across multiple classifiers; evaluated models on key metrics and selected the best-performing model for production deployment via FastAPI.
- Modularized data ingestion, training, and inference components into reusable pipeline stages, enabling reproducible experiments and easy maintenance.

Transformer-Based Sentiment Analysis (IMDB Reviews) — *Python, PyTorch, Hugging Face, BERT, Streamlit*

- Built an end-to-end sentiment classification system using BERT; implemented data ingestion, tokenization, fine-tuning, and evaluation pipelines with Hugging Face and PyTorch.
- Evaluated model performance using accuracy, precision, recall, and F1-score across positive and negative review classes on the IMDB benchmark dataset.
- Deployed the fine-tuned model with Streamlit for real-time inference, providing an interactive interface with a modular and production-ready project structure.

Education

Scaler School of Technology — 2023 – 2027
Undergraduate Program in Computer Science — Bangalore, India

- CGR: 6.07

Birla Institute of Technology and Science (BITS) — 2023 – 2027
B.Sc. (Hons.) in Computer Science — Bangalore, India

- CGPA: 6.85

Technical Skills

Languages	Python, Java, SQL, MongoDB
ML & Data Science	Supervised & Unsupervised Learning, Classification, Regression, Clustering, Feature Engineering, EDA, Model Optimization
Deep Learning & NLP	Neural Networks, CNNs, RNNs, LSTM, Transformers (BERT), Transfer Learning, Tokenization, TF-IDF, Word Embeddings, Sentiment Analysis, Text Classification
RAG	Vector RAG, Corrective RAG, Graph RAG, LangChain, FAISS, Groq, SPARQL, Gremlin, AWS Neptune
Computer Vision	YOLO (Ultralytics), OpenCV, Image Processing, Object Detection, Metadata Extraction
MLOps & Deployment	MLflow, FastAPI, Flask, Streamlit, Docker, Experiment Tracking, Pipeline Modularization
Cloud & Databases	AWS Neptune, Neo4j, OpenSearch, MongoDB, SQL